AMRITSAR GROUP OF COLLEGES NAAC Grade "A" 3rd Cycle | Autonomous College under Autonomous Category | (Since 2014) Conferred by UGC	AGCS-21704 Major Project Lab	Department of Computer Science and Engineering

B.Tech (CSE) 7th Sem – 3 (Group G1)

FORTNIGHTLY PROJECT PROGRESS REPORT-5

Date : 16-09-2026 to 30-09-2026
Group number : 07

Project Title : Nexora – An Intelligent Knowledge Workspace

Project Members:

S.No	Univ. Roll No	Name	Role
1	2333428	Shivam Kumar	Member
2	2333429	Shivam Kumar Pandit	Member
3	2333436	Sonu Kamat	Syndicate Leader

Progress Summary	1. Implemented JSON Web Token (JWT) stateless user authentication system with PBKDF2/SHA-256 password hashing. 2. Developed Role-Based Access Control (RBAC) separating Admin and User privileges across backend endpoints. 3. Built multi-tenant private workspace isolation using user-specific ChromaDB collection namespaces. 4. Integrated browser-native Web Speech API into the frontend interface for real-time voice query input. 5. Developed Whisper API fallback handler for cross-browser voice recognition compatibility. 6. Developed /api/generate-podcast REST API endpoint in FastAPI backend for audio content creation. 7. Created multi-agent dialogue script generation engine for synthesizing 2-host audio podcasts (Host A: Technical Lead, Host B: Co-host). 8. Integrated Web Audio API / gTTS audio player in frontend dashboard with speed controls and scrubber. 9. Built interactive Studio Artifacts engine for dynamic mind-map rendering and structured HTML summary export. 10. Enhanced Single-Page Web Dashboard (index.html, app.js, styles.css) with voice input controls and podcast player UI.
Work In Progress	1. Production Docker containerization setup using multi-stage builds. 2. Writing docker-compose.yml for orchestrating backend API, vector database, and frontend static assets. 3. Setting up automated CI/CD pipeline using GitHub Actions workflows. 4. Conducting concurrency load testing and memory leak audits under high user traffic. 5. Preparing comprehensive final major project documentation, presentation slides, and user guide.

Challenges And Issues	1. Ensuring complete data isolation between different user workspaces in ChromaDB. 2. Handling audio synthesis streaming and browser auto-play restrictions across different web browsers. 3. Maintaining speech-to-text accuracy in noisy environments and handling non-standard accents. 4. Optimizing audio script generation prompt structure to ensure natural conversational flow between 2 hosts. 5. Preventing session token security vulnerabilities and handling token refresh mechanisms efficiently.